Cardiometabolic Risk Clustering and Surrogate Markers of Insulin Resistance in Adolescents

Namrata Chhabra; Sarah Chhabra; Ranjit Patil; Sahiba Kukreja
International Journal of Applied and Basic Medical Research · 2026 · PMC13544660 · DOI: 10.4103/ijabmr.ijabmr_115_26

Resumo (Abstract)

Background: Adolescent obesity is increasingly linked to early insulin resistance (IR) and clustering of cardiometabolic abnormalities, including impaired glucose regulation, dyslipidemia, inflammation, and liver dysfunction. Identifying at-risk adolescents remains challenging due to physiological variability and a lack of standardized definitions. Aim: The aim of this study was to identify simple and reliable surrogate markers of IR and associated cardiometabolic risk in overweight and obese adolescents. Materials and Methods: In this cross-sectional study, 182 adolescents aged 10–19 years underwent detailed anthropometric, clinical, and biochemical evaluation, including fasting glucose, insulin, lipid profile, alanine aminotransferase (ALT), uric acid, and high-sensitivity C-reactive protein (hs-CRP). IR was defined as homeostatic model assessment of IR (HOMA-IR) ≥2.5, and associations with demographic, anthropometric, lifestyle, dietary, clinical, and metabolic factors were analyzed. Results: IR was identified in 62.6% of participants and was more common among obese and pubertal adolescents. Central obesity was present not only in the overweight and obese groups but also in 11.2% of normal-weight adolescents, indicating metabolic risk beyond body mass index (BMI). Measures of central adiposity, particularly waist-to-height ratio (WHtR), showed stronger associations with IR than BMI. IR-positive adolescents demonstrated clustering of metabolic, inflammatory, and hepatic abnormalities, including elevated glucose, insulin, triglycerides, triglyceride: high-density lipoprotein cholesterol (TG: HDL-C) ratio, hs-CRP, uric acid, and ALT, with reduced HDL-C. Acanthosis nigricans, low physical activity, poor sleep, and increased screen time were also more frequent. Metabolic syndrome was present in 13.18% of participants. Conclusion: IR and cardiometabolic risk are highly prevalent in adolescents with obesity. WHtR, TG: HDL-C ratio, ALT, hs-CRP, and acanthosis nigricans may serve as practical surrogate markers for identifying adolescents with HOMA-IR-defined IR and supporting early preventive strategies.

Introduction

Childhood and adolescent obesity have become a major public health concern worldwide, with prevalence rising across both high-income and low- and middle-income countries.[1] In 2021, an estimated 174 million individuals aged 5–24 years were living with obesity globally, and this number may rise to nearly 360 million by 2050 if current trends continue.[2] This increasing burden reflects changing dietary patterns, reduced physical activity, and the broader effects of urbanization and socioeconomic transition.

According to the World Health Organization, obesity is defined as abnormal or excessive fat accumulation that may impair health. Although body mass index (BMI) remains the most widely used measure, it has important limitations in children and adolescents. It does not account for fat distribution and may fail to identify those with excess central adiposity, which is more closely linked to metabolic risk. Measures, such as waist circumference, waist-to-height ratio (WHtR), and waist-to-hip ratio (WHR), may therefore provide a more meaningful assessment of cardiometabolic risk.[3 4]

One of the most important metabolic consequences of adolescent obesity is insulin resistance (IR), a condition in which tissues respond less effectively to insulin. It is influenced by genetic, hormonal, and lifestyle factors and is closely linked to type 2 diabetes, dyslipidemia, hypertension, and nonalcoholic fatty liver disease.[4 5]

Despite its importance, assessment of IR in adolescents remains challenging. Puberty introduces physiological variability in insulin sensitivity, and standardized cutoffs for markers such as homeostatic model assessment of IR (HOMA-IR) are lacking. In addition, markers such as alanine aminotransferase (ALT) and high-sensitivity C-reactive protein (hs-CRP) are not uniformly validated in younger populations. These challenges are further compounded by limited region-specific data from India.

This study aims to identify surrogate markers of IR and associated cardiometabolic risk in overweight and obese Indian adolescents. It examines anthropometric indices, clinical signs, lipid profiles, uric acid, hs-CRP, and liver enzymes and explores relationships with family history and components of metabolic syndrome.

The study addresses the following questions:

Materials and Methods

Study design and setting

This cross-sectional study was carried out at a tertiary care teaching center in North India to identify surrogate markers of IR and associated cardiometabolic risk factors in overweight and obese adolescents.

The study protocol was approved by the Institutional Ethics Committee of the participating institution (Approval No. Patho/71/2020; dated February 12, 2020). All procedures were performed in accordance with the ethical standards of the responsible committee on human experimentation and with the Declaration of Helsinki, 1975, as revised in 2013. Written informed consent was obtained from parents or legal guardians, and assent was obtained from all participating adolescents before enrollment. Permission for data collection was also obtained from the relevant school authorities.

Sample size and sampling

The minimum required sample size was calculated as 150 usingEpi Info software version 7 (Centers for Disease Control and Prevention, Atlanta, GA, USA) based on a 95% confidence level and a 5% alpha error. To account for possible dropouts or incomplete data, the target sample size was increased to 180. A total of 182 adolescents were ultimately included in the study. A multistage random sampling approach was used. One urban and one rural area were selected initially. From each area, one school was chosen at random. Students aged 10–19 years were then randomly selected for screening from grades 6 to 12.

Participants

A total of 357 students were screened, of whom 182 met the eligibility criteria and were enrolled. Participants were categorized according to weight status as normal weight, overweight, or obese. They were also assessed according to sex, pubertal stage, and place of residence.

Inclusion criteria

Adolescents aged 10–19 years who were classified according to the World Health Organization (WHO) 2007 growth reference for BMI[6] and were in Tanner stages T2–T5 were eligible for inclusion.

Exclusion criteria

Adolescents with known genetic syndromes, secondary causes of obesity, chronic systemic or endocrine disorders, or those receiving medications known to affect metabolic status were excluded.

Data collection

Data were collected using a structured questionnaire that recorded demographic details, personal and family medical history, dietary habits, physical activity, sleep duration, screen time, and selected environmental exposures. Each participant then underwent clinical examination, anthropometric assessment, and biochemical evaluation.

Clinical examination

The clinical assessment included grading of acanthosis nigricans on a scale of 0–4 and evaluation for gynecomastia, acne, striae, and hirsutism. Blood pressure was measured using an automated device in accordance with standard recommendations.[7]

Anthropometric assessments

Anthropometric measurements included height, weight, BMI, waist circumference, hip circumference, WHR, and WHtR. Central obesity was defined as a waist circumference at or above the 90 th percentile and/or a WHtR >0.5.[3]

Biochemical assessments

After an overnight fast, venous blood samples were collected from all participants for biochemical analysis. The parameters assessed included fasting plasma glucose, fasting insulin, hs-CRP, serum uric acid, liver enzymes, and lipid profile.

The lipid profile included total cholesterol, low-density lipoprotein (LDL) cholesterol, high-density lipoprotein (HDL) cholesterol, triglycerides, LDL/HDL ratio, and TG/HDL ratio. Liver enzymes included ALT and aspartate aminotransferase. IR was estimated using the HOMA-IR, calculated from fasting glucose and fasting insulin values.

Definition of metabolic syndrome

Metabolic syndrome was defined according to the International Diabetes Federation (IDF) criteria for adolescents.[8] Participants were classified as having metabolic syndrome if central obesity was present along with any two or more of the following abnormalities: elevated triglycerides, low HDL cholesterol, elevated blood pressure, or impaired fasting glucose.

Statistical analysis

Data were analyzed using IBM SPSS Statistics, version 29.0 (IBM Corp., Armonk, NY, USA). Continuous variables were expressed as mean ± standard deviation, and categorical variables were presented as frequencies and percentages. The Shapiro–Wilk test was used to assess normality. Depending on data distribution, parametric or nonparametric tests were applied. Associations

between IR and anthropometric, biochemical, clinical, and lifestyle variables were assessed using group comparisons and Pearson or Spearman correlation analyses, as appropriate. As the objective of the study was to identify clinically relevant surrogate markers associated with IR rather than to develop an adjusted prediction model, multivariable regression analysis was not performed, and independent predictive value was not assessed. $P < 0.05$ was considered statistically significant. All variables were complete, and no data were missing for the analyses performed.

Communication of findings and data availability

Findings were communicated confidentially to participants and their parents or guardians. Adolescents identified as being at increased metabolic risk were offered individualized counseling and health education related to diet, physical activity, and lifestyle modification.

Results

A total of 182 adolescents aged 10–19 years were included in the study. Based on the WHO 2007 BMI reference, 71 participants were classified as normal weight, 56 as overweight, and 55 as obese. Subgroup analyses were also performed according to sex, pubertal stage, and place of residence. Overweight and obese adolescents were more commonly from urban areas, whereas most normal-weight participants were from rural settings. The distribution of males and females was nearly equal across BMI categories, and pubertal and postpubertal participants were similarly represented.

Anthropometric measures showed a clear stepwise increase across BMI categories. BMI, waist circumference, and WHtR were all significantly higher in obese adolescents than in overweight and normal-weight participants [Figure 1], indicating increasing central adiposity with rising weight status.

Central obesity was highly prevalent in the study population. Waist circumference above the 90 th percentile was observed in 93 of 182 participants (51.1%), while 99 of 182 (54.4%) had a WHtR >0.5. Both indices were elevated in 81 of 182 adolescents (44.5%). Central obesity, defined as waist circumference at or above the 90 th percentile and/or WHtR >0.5, was considerably more frequent in the overweight and obese groups than in the normal-weight group. Among obese participants, nearly 89% met at least one criterion for central obesity, and more than three-quarters met both criteria.

The prevalence of hyperglycemia and hypertension also increased with BMI category. Hyperglycemia was present in 3 of 71 normal-weight adolescents (4.2%) and in 10 of 55 obese adolescents (18.2%). Hypertension was observed in 4 of 71 normal-weight participants (5.6%) and in 10 of 55 obese participants (18.1%). This clustering of metabolic abnormalities was most evident in the obese group [Figure 2].

Mean fasting insulin and HOMA-IR values rose progressively with increasing BMI. Obese adolescents had the highest mean fasting insulin level (19.91 µIU/mL) and mean HOMA-IR value (4.46), compared with 9.21 µIU/mL and 2.66, respectively, among normal-weight adolescents. For the cohort as a whole, the mean fasting insulin level was 15.09 µIU/mL, and the mean HOMA-IR was 3.50 [Figure 2].

IR was identified in 114 of 182 participants (62.6%). It was most frequent among obese adolescents and was particularly common among pubertal females and those from urban settings. Among obese females, 92.8% were insulin resistant, and 82.2% of pubertal adolescents met criteria for IR.

The clinical and biochemical characteristics of adolescents with and without IR are summarized in Table 1a - c . Compared with those without IR, adolescents with IR had significantly higher BMI, waist circumference, hip circumference, WHtR, and WHR. They also had higher fasting insulin and HOMA-IR values. Among the biochemical variables, triglycerides, TG: HDL-C ratio, hs-CRP, and ALT were significantly higher in the insulin-resistant group.

The correlation analysis between HOMA-IR and selected variables is shown in Table 2 . HOMA-IR demonstrated strong positive correlations with fasting insulin (r = 0.792), BMI (r = 0.459), WHtR (ρ = 0.431), and hs-CRP (ρ = 0.477). In contrast, HDL-C, total cholesterol, and uric acid showed weak or nonsignificant correlations.

Lifestyle and clinical features also differed between adolescents with and without IR. Low physical activity was reported more often among insulin-resistant participants than among those without IR (66 of 114, 57.9% vs. 26 of 68, 38.2%; P = 0.013). Acanthosis nigricans was present in 67 of 114 insulin-resistant adolescents (58.8%) compared with 18 of 68 participants without IR (26.5%; P < 0.001). Sleep disturbance was reported in 16 of 114 insulin-resistant participants (14.0%) and in 4 of 68 participants without IR (5.9%). A positive family history of type 2 diabetes was noted in 48 of 114 insulin-resistant adolescents (42.1%) compared with 23 of 68 adolescents without IR (33.8%). Excessive screen time was common in both the groups, affecting 83 of 114 insulin-resistant participants (72.8%) and 48 of 68 participants without IR (70.6%).

Metabolic syndrome, as defined by the IDF criteria, was present in 24 of 182 participants (13.2%). Of these, 16 of 24 (66.7%) were obese, 7 of 24 (29.2%) were overweight, and 1 of 24 (4.2%) was normal weight. The number of metabolic syndrome components increased with BMI category. Among obese adolescents, 22 of 55 (40.0%) had two components and 16 of 55 (29.1%) had three or more components, whereas 56 of 71 normal-weight participants (78.9%) had no components. Females accounted for 19 of 24 cases of metabolic syndrome (79.2%). Higher proportions were also observed among pubertal adolescents (13 of 24, 54.2%) and urban participants (14 of 24, 58.3%).

Dietary preference was associated with the number of metabolic syndrome components. Among adolescents with three components of metabolic syndrome, 54.1% were nonvegetarian, and among those with two components, 51.4% were nonvegetarian. In contrast, vegetarians were more commonly represented among those with one component (70.2%) or no component (56.3%).

Discussion

This study demonstrates a substantial burden of IR and early cardiometabolic risk among Indian adolescents with excess weight. IR was identified in 62.6% of participants, and the risk profile extended beyond obesity defined by BMI alone to include central adiposity, dyslipidemia, inflammation, elevated liver enzymes, and acanthosis nigricans. Importantly, the results also suggest that a subset of normal-weight adolescents with central obesity may already carry metabolic risk, reinforcing concerns that BMI alone may underestimate adiposity-related risk in the youth.[3 9 10]

Anthropometric profile and central adiposity

The anthropometric gradient shown in Figure 1 demonstrates a clear stepwise increase in BMI, waist circumference, WHR, and WHtR from normal-weight to overweight and obese adolescents. This pattern indicates that worsening overall adiposity is accompanied by increasing central fat accumulation. Among these measures, waist circumference and WHtR are particularly relevant because they more closely reflect visceral adiposity and cardiometabolic vulnerability than BMI alone,

a finding that is consistent with earlier work in children and adolescents.[3 9]

Central obesity and the limitations of body mass index alone

The high prevalence of central obesity in the present cohort further emphasizes this point. More than half of participants had a WHtR >0.5, and nearly 89% of obese adolescents met at least one central-obesity criterion. The presence of central obesity even within the normal-weight group is clinically important because it aligns with the concept of metabolically obese normal-weight individuals, in whom normal BMI may mask excess visceral fat and increased metabolic risk.[10] This issue is especially relevant in South Asian populations, where adverse metabolic outcomes often occur at lower BMI thresholds and where central adiposity may provide a more informative risk signal than BMI alone.[3 11]

Hyperglycemia, hypertension, and progression of metabolic risk

Figure 2 shows that hyperglycemia and hypertension became progressively more common with increasing adiposity, with the highest prevalence in obese adolescents. This clustering of abnormalities supports the view that obesity in adolescence is not an isolated anthropometric state but part of a broader cardiometabolic disturbance. Similar increases in dysglycemia and blood pressure have been described in Indian adolescents with overweight and obesity.[12] Mechanistically, IR may contribute to elevated blood pressure through sympathetic activation, renal sodium retention, endothelial dysfunction, and adipose-derived inflammatory signaling, thereby intensifying the cardiometabolic burden associated with obesity.[13]

Insulin resistance across body mass index category, sex, puberty, and residence

The progressive rise in mean fasting insulin and mean HOMA-IR across BMI categories in Figure 2 further supports declining insulin sensitivity with increasing adiposity. The greater burden of IR among obese adolescents, pubertal participants, and females is biologically plausible. Puberty is characterized by transient physiological reductions in insulin sensitivity related to growth hormone and sex-steroid changes, and this may amplify the metabolic impact of excess adiposity.[14] The urban predominance of obesity in this study is also consistent with the broader effects of sedentary behavior, dietary transition, and urbanization on adolescent metabolic health; however, the observation that rural adolescents were not protected from risk suggests that these transitions are now affecting diverse settings.[15]

Homeostatic model assessment of insulin resistance-defined insulin resistance

In the present study, IR was defined using a HOMA-IR cutoff of 2.5, which is consistent with previously reported adolescent thresholds.[4 16 17] This operational definition allowed the study to compare adolescents with and without IR and to examine the clustering of anthropometric, biochemical, clinical, and lifestyle-related markers across these groups.

The observed higher burden of central adiposity, dyslipidemia, inflammatory markers, ALT elevation, and acanthosis nigricans among adolescents with HOMA-IR-defined IR supports the clinical relevance of this classification. However, HOMA-IR should be interpreted as a practical surrogate marker rather than a direct measure of insulin sensitivity. In adolescents, its interpretation may vary according to pubertal stage, sex, ethnicity, and adiposity. Therefore, the findings are best interpreted

as clinically relevant associations with HOMA-IR-defined IR.

The central role of adiposity and metabolic dysfunction in insulin resistance

The differences summarized in Table 1 reinforce the central role of adiposity and metabolic dysfunction in IR. Adolescents with IR had significantly higher BMI, waist circumference, hip circumference, WHtR, and WHR, indicating that both total adiposity and central adiposity contribute to risk. They also had higher fasting glucose, fasting insulin, triglycerides, very LDL, total cholesterol/HDL ratio, triglyceride/HDL ratio, hs-CRP, and ALT. Although HDL cholesterol was lower in the insulin-resistant group, several traditional lipid variables such as total cholesterol and LDL cholesterol showed weaker discriminatory value, suggesting that composite indices such as triglyceride/HDL ratio may be more informative for early metabolic risk stratification than isolated lipid values. These findings support previous reports identifying the triglyceride/HDL ratio as a useful and low-cost surrogate of IR in the youth.[18]

The correlation analysis in Table 2 provides additional insight into the hierarchy of markers associated with IR. As expected, fasting insulin showed the strongest correlation with HOMA-IR. Among anthropometric variables, BMI, waist circumference, WHtR, and WHR all showed significant positive associations, with WHtR performing particularly well. Among biochemical markers, hs-CRP showed a notable correlation, supporting the role of chronic low-grade inflammation in adolescent IR, while triglycerides and the triglyceride/HDL ratio reflected the dyslipidemic component of this process.[18 19] The significant but weaker correlations of ALT and AST suggest early hepatic involvement, which may represent emerging nonalcoholic fatty liver disease in insulin-resistant adolescents.[20]

Lifestyle factors, family history, and acanthosis nigricans

The lifestyle and clinical findings also add important practical value. Lower physical activity was more common among insulin-resistant adolescents, which is in agreement with the known relationship between sedentary behaviors, reduced energy expenditure, and impaired insulin sensitivity.[21] Sleep disturbance was also more frequent in the insulin-resistant group, consistent with evidence linking sleep disruption and sleep-disordered breathing to metabolic dysfunction in the youth.[22] A positive family history of type 2 diabetes was common among adolescents with IR, suggesting that inherited susceptibility may interact with obesity-related exposures. The markedly higher prevalence of acanthosis nigricans among insulin-resistant participants is especially important in resource-limited settings, because this visible clinical sign may help identify adolescents who warrant targeted metabolic assessment.[23]

Metabolic syndrome and dietary pattern

The burden of metabolic syndrome in this study was concentrated largely in obese, female, pubertal, and urban adolescents, indicating that risk factors tend to cluster in those already showing central adiposity and IR. The pattern of increasing metabolic syndrome components with higher BMI supports the progressive nature of cardiometabolic derangement during adolescence.[8 12] The observed tendency for nonvegetarian adolescents to have a greater burden of metabolic syndrome components, while exploratory, is also noteworthy. Prior work suggests that plant-based dietary patterns may improve insulin sensitivity and reduce cardiometabolic risk, whereas diets richer in saturated fat and lower in fiber may worsen metabolic outcomes.[24 25] Because dietary intake in

the present study was self-reported and not quantified in detail, this finding should be interpreted cautiously, but it supports the need to include dietary counseling in adolescent risk reduction strategies.

Collectively, the findings address the three research questions. First, anthropometric measures, particularly waist circumference and WHtR, were closely associated with IR and highlighted the limitations of BMI alone. Second, IR was accompanied by metabolic deviations including higher fasting insulin, triglycerides, TG: HDL-C ratio, hs-CRP, ALT, and greater clustering of metabolic syndrome components. Third, the broader cardiometabolic risk profile included central adiposity, dyslipidemia, inflammatory and hepatic markers, acanthosis nigricans, reduced physical activity, sleep disturbance, and family history of type 2 diabetes, supporting a practical screening approach that integrates anthropometric, biochemical, clinical, and lifestyle indicators.

Clinical implications, strengths, and limitations

While the relationship between central adiposity and IR is well recognized, the present study adds value by demonstrating how simple anthropometric, biochemical, clinical, and lifestyle markers cluster in a school-based Indian adolescent cohort and may be used pragmatically to identify adolescents requiring further metabolic assessment. Taken together, these findings support a screening approach that moves beyond BMI alone. The combination of central adiposity measures, especially WHtR; biochemical markers such as triglyceride/HDL ratio, hs-CRP, and ALT; and clinical indicators such as acanthosis nigricans may offer a practical framework for identifying adolescents at risk of IR and future cardiometabolic disease. A major strength of this study is the integration of anthropometric, biochemical, clinical, and lifestyle data in a cohort that included both urban and rural participants. However, the cross-sectional design limits causal and temporal inference; the sample was drawn from a single region of North India, and lifestyle variables, including physical activity, sleep patterns, and dietary habits, were self-reported and may therefore be affected by recall and reporting bias. Furthermore, school-specific factors, including socioeconomic background, cafeteria food options, physical activity opportunities, academic routines, and health awareness, may have influenced the observed lifestyle and metabolic risk patterns. Although multivariable regression was not performed, the observed group differences and correlations identify clinically relevant markers associated with IR and provide a basis for future validation in larger cohorts. Longitudinal studies are needed to determine how these markers track over time and whether targeted interventions based on these simple screening tools can reduce later risks of type 2 diabetes, nonalcoholic fatty liver disease, and cardiovascular disease.

Conclusion

This study demonstrates a significant burden of IR, central adiposity, and early metabolic risk among overweight and obese Indian adolescents, with a particularly high prevalence among pubertal females and urban residents. Anthropometric measures such as WHtR, clinical indicators such as acanthosis nigricans, and HOMA-IR-based classification offer accessible and practical tools for early identification of at-risk individuals. These findings emphasize the urgent need for adolescent-focused screening and prevention initiatives, particularly within school and community health frameworks, to reduce long-term risks of diabetes, nonalcoholic fatty liver disease, and cardiovascular complications.

Conflicts of interest

There are no conflicts of interest.